Climate Informatics 2026 (2026), xx: 1–2
doi:10.1017/xxxx

CAMBRIDGE UNIVERSITY PRESS

CLIMATE INFORMATICS 2026
EXTENDED ABSTRACT

Exploring Adversarial Attacks in AI Weather Models for Generation of High-resolution Tropical Cyclones

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Received xx xxx xxxx

Keywords: ai weather model, adversarial example, tropical cyclone

1. Introduction and objectives

There has been recent interest in the advantage of differentiability of AI-weather models to enable direct computation of model sensitivities to initial conditions. In the field of machine learning, adversarial attacks leverage these sensitivities to influence the output of the prediction system by finding optimal initial condition perturbations. In weather forecasting, this methodology can be seen under two lenses: differentiable models are susceptible to malicious attacks aimed at distorting operational forecasts [1], while having access to sensitivities is an opportunity to further our understanding of real events through the generation of synthetic forecasts. Adversarial examples - perturbed initial conditions obtained from adversarial attacks - have been used in [2] to create even more extreme forecasts of a heatwave, suggesting a storyline approach to understanding black-swan heatwave events.

This work further explores this storyline approach by considering adversarial attacks on tropical cyclone predictions at high resolution and in an operational setting.

2. Methods and data

We use Operational GraphCast [3] and the fine-tuned checkpoint created by [4] to adversarially attack tropical cyclone forecasts at 0.25° resolution, initialized from IFS HRES analysis. For selected events, we aim to find adversarial examples to deviate tropical cyclone track at increasing distances from the ground truth. We constrain perturbations to the second input time-frame, and to non-forcing and non-static variables. To limit adversarial example perturbation magnitudes, we introduce a L_∞ penalty term to each objective, approximated by a softmax over the spatial grid in normalized space and weighted by variable and pressure level. We then examine perturbation magnitudes in relation to the track deviation distances. Adversarially created forecasts are further evaluated using spherical harmonic power spectra to assess physical consistency and the interpretability of perturbations are tested by transferring initial inputs to other AI-weather models.

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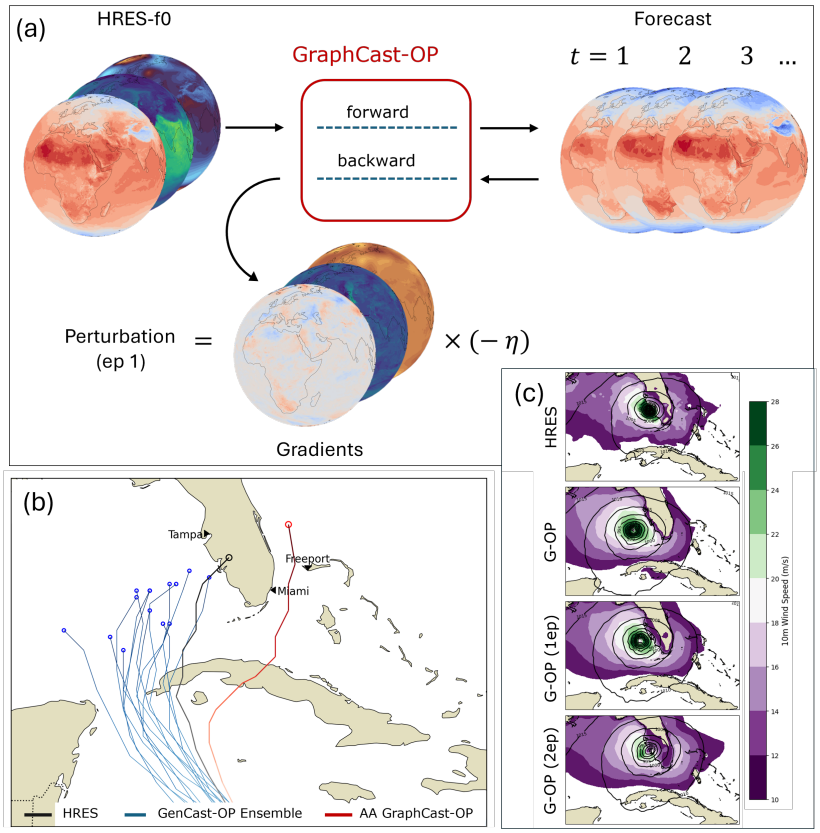

Figure 1. *(a) Schematic of a single optimisation step in an adversarial attack on GraphCast-OP. The backward pass computes the gradient of the loss with respect to the full inputs, yielding a perturbation proportional to the negative gradients, for learning-rate η. (b) Example of an adversarial attack deviating the track of Hurricane Ian beyond the uncertainty modeled by a GenCast-OP ensemble. (c) Evidence of enhanced "blurring" when adversarially attacking a forecast of Hurricane Ian.*

3. Preliminary results and/or expected outcomes

We show that adversarial examples for deviating tropical cyclones to various targets are achievable with few optimization epochs, extending recent literature to 0.25° resolution. Furthermore, preliminary findings show that optimizing for more complex objectives leads the model to require larger perturbation magnitudes, while forecasts exhibit increasingly nonphysical spherical harmonic power spectra. Encouragingly, optimizing for clearly nonphysical objectives is difficult, with the model finding nonphysical solutions to the optimization problem. This work then contributes evidence in favor of AI-weather storyline approaches for operational forecasting services in extreme events unaccounted for by under-dispersed ensemble forecasting systems.

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